



UNIVERSITY COLLEGE OF ENGINEERING SCIENCE AND TECHNOLOGY HYDERABAD
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MID EXAMINATION ANSWER BOOKLET

H.T. No.

2	3	0	1	1	M	2	1	1	2
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NAME OF THE EXAMINATION

Btech 2nd year 2nd sem 1st mid

SUBJECT DBMS

BRANCH/SPECIALIZATION

CSE-IDDMP

DATE OF EXAM:

6/3/2024

Signature of Invigilator

Please tick (✓) the questions and bits answered by you
in the space provided above the dotted line

	1	2	3	4	5	6	7	8	9	10
Part A	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

Part B

Question No.	1	2	3	4	5	6	7	8
Tick Marks	a							
Tick Marks	b							
Tick Marks	c							
Question Wise Total Marks								
(Part A + Part B) = Grand Total (in Figures)								
in words								

INSTRUCTIONS TO CANDIDATES OVERLEAF

Part - B

1) Database is defined as the collection of interrelated data stored in the form of tables.

→ When we used the traditional file system we faced many problems like

- (i) Data redundancy
- (ii) Access difficulty
- (iii) Data isolation
- (iv) Integrity
- (v) Atomicity
- (vi) concurrency
- (vii) security issues

→ To overcome all these problems Database management systems were introduced and they solve all these problems due to their unique characteristics and they are:-

→ database approach uses tables to store interrelated data with concepts like primary, Key Foreign Key data redundancy is handled as there cannot be duplicate values in the primary Key.

→ As we store the data in form of tables with keys linking the data, accessing the data also became very easy compared to the traditional file management systems.

→ To tackle data isolation we load the data from the datasets of different file types and load the data into the tables which solves the data isolation problem and the ease of access does not make the data isolate.

→ The data stored in the database is consistent and reliable as the database has a lock system, when a transaction is occurring the database locks and either fully execute or not execute without ^{leaving} being in the intermediate steps making the data consistent.

→ The security of the database is also very high as it has levels of abstraction, only the database administrator has the full access to the database making it a secure method.

→ The schema, the architecture of the database is done such that it is able to tackle all the major problems like data redundancy, concurrency, atomicity, integrity, security, data isolation and access difficulty.

2) There are six main fundamental operations of Relational algebra and they are:-

- (i) SELECT
- (ii) PROJECT
- (iii) UNION
- (iv) SET DIFFERENCE
- (v) CARTESIAN PRODUCT
- (vi) RENAME

(i) SELECT:- consider a relation 'r' then the SELECT operation is represented as $\sigma_p(r)$ where p is the selection predicate.

→ Select is defined as

$$\sigma_p(r) = \{t / t \in r \text{ and } p(t)\}$$

where p is the formula in propositional calculus consisting of terms connected by $\wedge, \vee$ etc:-

→ $\langle \text{attribute} \rangle \text{ op } \langle \text{attribute} \rangle$ or $\langle \text{cond} \rangle$

where $\text{op} = \leq, \geq, =, >, <$ etc:-

(ii) PROJECT:- Project operation is represented as $\Pi_{A_1, A_2, A_3, \dots, A_k}(r)$ where A_i is the attribute and r is the relation.

→ the result is defined as relation of k columns obtained by erasing columns ~~are~~ that are not listed in the condition of the operator.

→ duplicate rows are removed from the result after the operation is applied and the result tends to be a column.

(iii) UNION:- Union operation is represented as $r \cup s$ where r and s are relations.

→ union is defined as

$$r \cup s = \{t \mid t \in r \text{ or } t \in s\}$$

→ where r, s are two relations and r and s should have the same arity. i.e both the relations should have same number of attributes.

→ The type of attributes must be compatible with each other.

(iv) SET DIFFERENCE:- Set difference operation is represented as $r - s$ where r and s are two comparable relations.

→ It is defined as

$$r - s = \{t \mid t \in r \text{ and } t \notin s\}$$

→ Here both r and s relations should have same arity.

→ Attribute domains of both the relations must be compatible.

(v) CARTESIAN PRODUCT:- Cartesian product operation is represented as $r \times s$ where r and s are two relations.

→ Cartesian Product is defined as

$$R \times S = \{t \mid t \in R \text{ and } t \notin S\}$$

$$R \times S = \{tq \mid t \in R \text{ and } q \in S\}$$

→ assume the attributes of $r(R)$ and $s(S)$ are disjoint i.e. $(R \cap S) = \emptyset$

→ if not disjoint renaming must be used.

(vi) RENAME:- This Rename operation allows us to name, refer to relational algebra expression. It is represented as $\rho_x(E)$

→ It refers one relation by more than one name.

→ If a relational algebra expression E has an arity n then $\rho_{(A_1, A_2, \dots, A_n)}(E)$ returns result of expression E under the name X , with attributes renamed to $A_1, A_2, \dots, A_n$

3) E-R model is known as Entity relationship model has an entity in the model which can be a person, a place anything

→ In the above entity relation diagram we can see that all the entities are enclosed in , all the attributes are enclosed in  and all the relations

→ We can see a weak entity relation diagram in the center of the diagram as there is no key in that part

→ We can entities and relationships branching out from the entity customs and with the help of relationships we are able to access all the data in a structured manner with the help of this diagram.

6)

(i) DDL:- DDL is known as the data definition Language in the Structured Query Language where keywords are used to define things

→ Operations like

CREATE, ALTER, DROP, TRUNCATE

are in the category of Data definition language as the command CREATE, creates a table where we are defining a table.

→ DML - DML is known as the data Manipulation Language in the Structured Query Language where keywords are used to manipulate the given data.

→ Operations like

SELECT, INSERT, UPDATE

are in the category of Data manipulation language as the SELECT operation is used to present or manipulate the data.

→ We use these DML operations on the data and find useful insights from the dataset.

(iii) Database Schema:- Database Schema is the main core structure of the database and also a vital part of the architecture of Database management system.

→ It consists of 3 levels

(a) Physical Level Schema:- In this level we can understand how data is stored and the memory allocation.

Logical Level Schema:- In this level we can see the relationships and the logic of the database.

View Level Schema:- In this level of schema we can see the different views from different perspectives as we are interpreting the view level schema.

→ All of these consists come altogether and form the database schema which form the core structure of the database.